Lecture 3: Mean-Variance Tradeoff, Method of Moment Estimator

IEOR E4709: Statistic Analysis and Time Series
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1 Bias-Variance Tradeoff

Let $\hat{\theta}$ be an estimator of the (unknown) parameter θ. The *mean squared error* of the estimator $\hat{\theta}$ is

$$MSE(\hat{\theta}) := \mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right].$$

Result: $\mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right] = Var\left(\hat{\theta}\right) + \left(Bias(\hat{\theta})\right)^2$. To see this, notice that

$$\begin{aligned}
\mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right] &= \mathbb{E}\left[\left(\hat{\theta}-\mathbb{E}\left[\hat{\theta}\right]+\mathbb{E}\left[\hat{\theta}\right]-\theta\right)^2\right] \\
&= Var\left(\hat{\theta}\right) + (Bias(\hat{\theta}))^2 + 2\mathbb{E}\left[\left(\hat{\theta}-\mathbb{E}\left[\hat{\theta}\right]\right)\left(\mathbb{E}\left[\hat{\theta}\right]-\theta\right)\right] \\
&= Var\left(\hat{\theta}\right) + (Bias(\hat{\theta}))^2
\end{aligned}$$

Since the MSE decomposes into a sum of the bias and variance of the estimator, both quantities need to be as small as possible to achieve a good estimation performance. Typically, one trades off some increase in bias for a decrease in variance, and vice-versa. Assume we have a sequence of i.i.d.. random variables $X_1, X_2, \ldots, X_n$, where each X_i has mean μ and variance σ^2. Why do we estimate the population mean using the sample mean $\bar{X}$ as opposed using a single random variable X_1? Because there is less variability in averages than in individual observations. Indeed, both $\bar{X}$ and X_1 are unbiased estimators of the population mean μ. However, $MSE(\bar{X}) = \frac{\sigma^2}{n} \leq \sigma^2 := MSE(X_1)$.

Let us next provide an example to show that different unbiased estimators may have different mean-squared errors.

Example: Consider the uniform distribution on the interval $(0, 1)$. The sample mean estimator is $\bar{U} = \frac{1}{n}\sum_{i=1}^{n} U_i$. We know from our previous arguments that this is an unbiased estimator of the

mean $\frac{1}{2}$. The variance of $\bar{U}$ is $Var(\bar{U}) = \frac{1}{n}Var(U_1)$. The variance of the uniform distribution in the interval (a, b) is $\frac{(b-a)^3}{12}$. So, in our specific case, it is 1/12. Hence, $MSE(\bar{U}) = \frac{1}{12n}$.

Consider now the estimator $\hat{\mu} := \frac{n+1}{2n} \max_{1 \leq i \leq n} U_i$. Notice that the maximum of n uniform random variables $U_1, U_2, \ldots, U_n$ has mean given by $\frac{n}{n+1}$ and standard deviation $\frac{1}{n+1}\sqrt{n/(n+2)}$.[1] Then $\mathbb{E}[\hat{\mu}] = \frac{n+1}{2n}\frac{n}{n+1} = \frac{1}{2}$. So, $\hat{\mu}$ is also an unbiased estimator of the mean of the uniform distribution. We have that $MSE(\hat{\mu}) = \frac{(n+1)^2}{4n^2}\frac{1}{(n+1)^2}\frac{n}{n+2} = \frac{1}{4n(n+2)}$. Comparing the MSE of $\bar{U}$ and of $\hat{\mu}$, we have that $MSE(\hat{\mu}) \leq MSE(\bar{U})$ if and only if $n > 1$. Hence, the nonlinear estimator $\hat{\mu}$ has a smaller mean squared error than the sample mean estimator $\bar{U}$. It is, however, well known that the sample mean estimator has the smallest MSE among all unbiased *linear estimators*.

2 Method of Moments

The method of moments is a parameter estimation technique which is only based on the law of large numbers. Let us first give a general description of the technique, and then provide an example. Consider independent and identically distributed random variables $X_1, X_2, \ldots$ chosen according to a parametric probability distribution depending on the parameter θ. Let f be a real valued function. If $k(\theta) = \mathbb{E}_\theta[f(X_1)]$, where $\mathbb{E}_\theta$ indicates that the expectation is being taken with respect to the probability distribution parameterized by θ, then

$$\frac{1}{n}\sum_{i=1}^{n} f(X_i) \to k(\theta), \tag{1}$$

with probability one, as $n \to \infty$, by the strong law of large numbers. The method of moments amounts to equating the population moment to the sample moment estimates. Concretely, it goes as follows. The estimation procedure consists of four steps. Denote by $\mu_m := \mathbb{E}[X^m] = k_m(\theta)$ the m-th moment. Suppose that the model has d parameters, i.e. $\theta = (\theta_1, \theta_2, \ldots, \theta_d)$.

- **Step 1.** Obtain the expressions $k_m(\theta)$ for the first d moments, i.e.,

  $$\mu_1 = k_1(\theta_1, \theta_2, \ldots, \theta_d), \qquad \mu_2 = k_2(\theta_1, \theta_2, \ldots, \theta_d), \qquad \ldots \qquad \mu_d = k_d(\theta_1, \theta_2, \ldots, \theta_d),$$

  yielding d equations in d unknowns (the unknowns being the parameters $\theta_1, \theta_2, \ldots, \theta_d$).

- **Step 2.** We solve for the d parameters as a function of the moments, i.e.,

  $$\theta_1 = g_1(\mu_1, \ldots, \mu_d), \qquad \theta_2 = g_2(\mu_1, \ldots, \mu_d), \qquad \ldots, \qquad \theta_d = g_d(\mu_1, \ldots, \mu_d) \tag{2}$$

- **Step 3.** Based on the samples $X := (X_1, X_2, \ldots, X_n)$, compute the first d sample moments

  $$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \overline{X^2} = \frac{1}{n}\sum_{i=1}^{n} X_i^2, \qquad \ldots \qquad \overline{X^d} = \frac{1}{n}\sum_{i=1}^{n} X_i^d.$$

  By the strong law of large numbers, for each moment, $m = 1, \ldots, d$, we have that $\mu_m \approx \overline{X^m}$.

[1] $\mathbb{P}(\max_{i=1,\ldots,n} U_i <= x) = \mathbb{P}(U_1 \leq x, U_2 \leq x, \ldots U_n \leq x) = x^n$. Hence, the probability density is $f(x) = \frac{\partial \mathbb{P}(\max_{i=1,\ldots,n} U_i <= x)}{\partial x} = nx^{n-1}$. The mean is then equal to $\int_0^1 nx^{n-1}xdx = \frac{n}{n+1}$. For the standard deviation, verify it yourself with a direct calculation.

- **Step 4.** Replace the distributional moments μ_m by the sample moments $\overline{X^m}$. Then the equations in (2) give us formulas for the estimators $(\hat{\theta}_1, \ldots, \hat{\theta}_d)$. Calculating these estimators on the samples X, we obtain the estimates

$$\hat{\theta}_1(X) = g_1(\bar{X}, \overline{X^2}, \ldots, \overline{X^d}), \hat{\theta}_2(X) = g_2(\bar{X}, \overline{X^2}, \ldots, \overline{X^d}), \ldots\ldots, \hat{\theta}_d(X) = g_d(\bar{X}, \overline{X^2}, \ldots, \overline{X^d}).$$

Example: Let $X_1, X_2, \ldots, X_n$ be a random sample of exponential distributed random variables with density

$$f_\beta(x) = \beta e^{-\beta x}, \qquad x > 0.$$

The cumulative distribution function is

$$F(x) = 1 - e^{-\beta x}, \qquad x > 0.$$

How many parameters do we need to estimate, i.e., what is d? In this case, $d = 1$, as we only need to estimate β. Hence, in the first step of the above procedure, we compute $\mu = k_1(\beta) = \frac{1}{\beta}$. In step 2, we solve for β as a function of μ. This yields $\beta = g(\mu) = \frac{1}{\mu}$. In step 4, we obtain $\hat{\beta} = \frac{1}{\bar{X}}$, i.e., replacing the (unknown) population mean with its sample mean.